

Project Revision Notes — Bullet Points

1. Project kya hai?

- AI/NLP-based **Named Entity Recognition (NER) System**
- Input text se important entities identify karta hai.
- Entities ko categories/labels deta hai.

2. NER kya hai?

- **NER = Named Entity Recognition**
- Text mein se important entities identify aur classify karta hai.
- Examples:
 - Person
 - Location
 - Organization
 - Date

3. Example

Input:

Rahul 15 August ko Delhi gaya.

Output:

- Rahul → PERSON
- Delhi → LOCATION
- 15 August → DATE

4. Project kyu bana rahe hain?

- Large text ko manually analyze karna time-consuming hai.
- NER automatically important information extract karta hai.
- Manual effort aur time reduce hota hai.
- Unstructured text ko structured information mein convert karta hai.

5. NLP kya hai?

- **NLP = Natural Language Processing**
- AI ka field jo computer ko human language:
 - Read
 - Understand
 - Process
 - Analyze
 karne mein help karta hai.

6. NLP project mein kyu use ho raha hai?

- Hamara input human language/text hai.
- Hume text ka meaning aur important entities identify karni hain.
- Isliye NLP techniques/models use hote hain.

7. User flow

- User text enter karega.
- Text NER model ko jayega.
- Model entities identify karega.
- Har entity ko label milega.
- Final result user ko display hoga.

8. Possible users

- Researchers
- Students
- Developers
- Companies/Organizations
- Educational institutions

9. Annotation kya hai?

- Text ke kisi important part ko **label dena** annotation kehlata hai.
- Example:

- Rahul → PERSON
- Delhi → LOCATION

10. Annotated dataset kya hai?

- Bahut saare labelled examples ka collection.
- Example:
 - Rahul → PERSON
 - Delhi → LOCATION
 - Microsoft → ORGANIZATION

11. Annotated dataset kyu important hai?

- Model ko examples se learn karna hota hai.
- **Input + Correct Label** model ko pattern sikhata hai.
- Supervised learning mein labelled data important hota hai.

12. Model training

- Annotated dataset model ko diya jata hai.
- Model patterns learn karta hai.
- Baad mein naye/unseen text par prediction karta hai.

13. Fine-tuning

- Pehle se trained model ko specific dataset/task par further train karna.
- Example:
 - Pretrained model
 - ↓
 - Hindi NER dataset
 - ↓
 - Fine-tuned NER model

14. LLM ka role

- LLM context samajhne mein help kar sakta hai.

- Entities aur labels suggest kar sakta hai.
- Future mein AI-assisted entity recognition ke liye use ho sakta hai.

15. Current objective

- NER module develop karna.
- Person, Location, Organization, Date etc. identify karna.
- Model ki performance evaluate karna.

16. Model evaluation metrics

- **Precision:** Predicted entities mein kitni correct hain.
- **Recall:** Actual entities mein se kitni identify hui.
- **F1-score:** Precision aur Recall ka balanced measure.

17. F1-score kyu important hai?

- Precision aur Recall dono ko consider karta hai.
- Sirf accuracy kabhi-kabhi misleading ho sakti hai.
- NER evaluation ke liye useful metric hai.

Future Goals

18. Question & Answer Generation

- User paragraph dega.
- System relevant questions generate karega.
- Corresponding answers generate karega.
- Answer verification check karega ki answer paragraph ke context se supported hai.

19. Sentiment Analysis

- Textual opinion/review analyze karega.
- **1–5 rating** predict karega.
- Overall sentiment identify karega:
 - Positive

- Negative
- Neutral

20. Mood/Emotion Analysis

- Text se emotion identify karne ki future capability.
- Examples:
 - Happy
 - Sad
 - Angry
 - Frustrated
 - Satisfied

21. Future Integration

- Existing NER system
- - Question & Answer Generation
- - Sentiment & Mood Analysis
- = **Unified AI-based Text Analysis Platform**

Objective vs Future Goal

Objective

- Jo **abhi/current project mein implement** kar rahe hain.
- Hamare case mein → **NER System**

Future Goal

- Jo future mein add/improve karna hai.
- Hamare case mein:
 - Q&A Generation

- Sentiment Analysis
 - Mood Analysis
 - Complete platform integration
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SWOT Analysis

Strengths

- Important information automatically extract karta hai.
- Manual effort reduce karta hai.
- Large text analysis mein useful.
- Future mein easily expandable.

Weaknesses

- Accuracy dataset/model quality par depend karti hai.
- Ambiguous text mein errors aa sakte hain.
- Different domains/languages mein performance vary ho sakti hai.

Opportunities

- More languages add kar sakte hain.
- Q&A generation add kar sakte hain.
- Sentiment/Mood analysis add kar sakte hain.
- Complete AI NLP platform bana sakte hain.

Threats

- Incorrect predictions.
 - Biased training data.
 - Privacy issues with sensitive text.
 - Existing advanced NLP tools/models.
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1-Line Project Summary

Hum ek AI/NLP-based NER system bana rahe hain jo user ke input text se important entities jaise Person, Location, Organization aur Date ko automatically identify aur classify karta hai. Future mein ise Question & Answer Generation aur Sentiment & Mood Analysis ke saath integrate karke ek complete AI-based text analysis platform banaya jayega.